

Jawaharlal Nehru Govt. Engineering College Sundernagar, Mandi (H.P.)

Course Name: Data Warehousing and Data Mining Course Code: CS-111 Examination: 1st Periodical Examination

Scheme: NEP Branch: CSE (AI & ML) Semester: V

Question Paper for 1st Periodical Examination (September 2025) (Session August-Dec 2025)

Time: One and a half hours

Max. Marks: 30

Note: Attempt all questions.

Course Outcomes:

CO1. Study data warehouse principles and its working.

CO2. Learn data mining concepts and understand Association Rule Mining.

CO3. Study Classification Algorithms.

CO4. Gain knowledge of how data is grouped using clustering techniques.

Q.1 Answer the following questions:

(5x1=5)

- (a) Define data model.
- (b) Briefly explain the fact-less-fact table?
- (c) What type of data do a fact table and dimension table store?
- (d) What does MOLAP stand for?
- (e) Define a data cube in the context of data warehousing.

Q.2 Compare and contrast operational database systems and data warehouses in terms of purpose, data type, and usage.

(5)

Q.3 Describe the architecture of a data warehouse. Explain the role of each key component.

(5)

Q.4 What is ETL (Extraction-Transformation-Loading) in data warehousing? Describe its process and importance.

(5)

Q.5 Answer the following questions:

(5x2=10)

- (a) Define data mining and discuss its primary functionalities, illustrating each with appropriate examples.
- (b) Explain the concept of a schema. Differentiate star schema and snowflake schema.